

Check 43 later

Others: (9 Marks) Section 1: OS Fundamentals and Concurrent Execution.....Each Question 1/2 Mark

1) A Subroutine in the operating system to process a particular type of interrupt is called _____	a) Semaphore b) Dispatcher c) Bootstrap d) <u>Non-of-the-above</u>
2) A _____ move from one state to another state during its life time	a) Swapper b) Pager c) <u>Process</u> d) I-node
3) The one program that is running at all times on the computer is _____	a) <u>Kernel</u> b) Bootstraper c) Semaphore d) Non-of_the_above
4) Internal memory fragmentation	a) occurs in Demand paging technique b) occurs in contiguous memory technique c) decreases memory utilization. ✓ d) <u>All-Of-The-Above</u>
5) occurs if a low priority job, though ready, is never scheduled to run because there is always a higher priority job ready.	a) Priority Inversion Problem. b) Multi-levels feedback queues. c) deadlock d) <u>Starvation</u>
6) The busy waiting means:	a) Process blocked waiting for I/O to complete. b) Process spent its quantum for other process. c) Trying to connect with device which doesn't respond. d) <u>None of the above.</u>
7) A semaphore ____.	a) <u>is essentially an integer variable</u> b) is accessed through only one standard operation c) can be modified simultaneously by multiple threads d) cannot be used to control access to a thread's critical sections
8) Blocking implementation of the semaphore mechanism needs _____	a) Swapping b) Paging c) SMP d) <u>waiting queue</u>
9) In Peterson's solution, _____ condition(s) to avoid race conditions are satisfied.	a) one b) two c) <u>three</u> d) four
10) Counting semaphores are used for solving ____.	a) mutual exclusion b) Priority inversion problem c) Starvation d) <u>access control problem to a resource having finite number of instance</u>
11) A deadlocked state occurs whenever ____.	a) a process is waiting for I/O to a device that does not exist b) the system has no available free resources

	c) <u>every process in a set is waiting for an event that can only be caused by another process in the set</u> d) a process is unable to release its request for a resource after use
--	---

12) Race Condition can be avoided if only mutual exclusion is granted.	F
13) Semaphores implemented with busy waiting can be useful in SMP environment.	T
14) A Memory Management Unit (MMU) is responsible for mapping the logical addresses produced by processes into <u>physical</u> addresses.	T ✓
15) Demand paging is a technique used by memory manager to avoid internal fragmentation.	F ✗
16) In <u>paging</u> memory management technique page size can differ from frame size.	F ✗
17) Virtual memory makes programming easier because programmers don't have to worry about the amount of physical memory that will be available run their programs.	T ✓
18) TLB can improve the effective memory access time using contiguous memory allocation technique	F ✗

CLO 6: (6 Marks) Section 2: Memory Management

Each Question 1/2 Mark

19) In <u>contiguous</u> memory allocation with <u>variable size</u> <u>partitions</u> <u>external fragmentation will not occur when</u> _____	a) first fit is used b) best fit is used c) worst fit is used d) <u>no matter which algorithm is used, it will always occur</u>
20) A <u>process</u> is _____ if it spends more time paging and doing page faults than executing.	a) Segmenting b) Thrashing c) Paging d) swapping
21) The address of a segmentation table in memory is pointed by _____	a) stack pointer b) page table base register c) page register d) <u>None of the above</u>
22) In contiguous memory allocation with <u>fixed sized</u> <u>partition</u> , the degree of multi programming is bounded by _____	a) the number of partitions b) the CPU utilization c) the memory size d) all of the above
23) Given the reference string of page accesses: <u>1 2 3 4 2 3 4 1 2 1 1</u> and a system with three page frames, what is the final configuration of the three frames after the true LRU algorithm is applied?	a) 2, 1, 3 b) <u>4, 1, 2</u> ✓ c) 3, 4, 1 d) 1, 2, 3
24) In a 16-bit machine with page size <u>1K</u> byte and configured with maximum adequate frames. Which program size is impossible to load into memory managed by paging technique? <u>2¹⁶ = 65,536 / 1024 = 64 Kbyte</u>	a) 16 KB b) 5.5 KB c) 32000 Bytes d) 68 KB ✗ <u>32 Kbytes</u> <u>512 Kbytes</u>
25) What is compaction?	a) a technique for overcoming internal fragmentation

	b) a paging technique c) <u>a technique for overcoming external fragmentation</u> ✓ d) a technique for overcoming fatal error
26) In a 16-bit machine with <u>4k</u> page size. What is the maximum number of pages that can be addressed in this system? $2^{16} = 64K/4 = 16$	a) 16 b) 4 c) <u>8</u> d) 32
27) If a page number is not found in the TLB, then it is <u>known as a</u> :	a) <u>TLB miss</u> ✓ b) Buffer miss c) TLB hit d) All of the mentioned
28) In Paging Technique every address generated by the CPU is divided into two parts :	a) frame bit & page number b) <u>page number & page offset</u> c) page offset & frame bit d) frame offset & page offset
29) At certain moment, the memory holes was <u>100K</u> , <u>500k</u> , <u>200k</u> and <u>600k</u> (in memory order) after allocating processes of <u>210 k</u> , <u>517 k</u> , <u>110 k</u> and <u>160 k</u> respectively using <u>first fit algorithm</u> , the remaining holes in the memory order are:	a) 100 k, 180 K, 200 k, 600 K b) 100 K, 276 K, 80 K, 183 K c) <u>100 K, 20 K, 200 K, 83 K</u> d) 100 K, 180 K, 20 K, 83 K

CLO 8: (Marks 3) Section 3: Disk Fundamentals

Each Question has 1/2 Mark

30) File Control Block contains the file size beside other file attributes.	T
31) Disk <u>Linked</u> allocation method ensures that only one access is needed to get a disk block using <u>direct</u> access.	F
32) File whose blocks were allocated using Indexed allocation method needs <u>2 disk block</u> access in order to read or write any byte of the file.	T ✓
33) On 32-bit machine a FAT file system can't be used to format a terabyte hard disk as one partition.	T ✓
34) Which of the following cannot be considered as disk access delay?	a) transfer delay ✓ b) rotational delay ✓ c) seek delay ✓ d) <u>cache memory access delay.</u>
35) Disk file system with contiguous allocation is suitable for formatting?	a) <u>CDROM</u> ✓ b) Video tape c) Hard Disk d) Flash memory sticks

~~Others:~~ (3 Marks) Section 4: Process Scheduling

~~36)~~ What would happen if round robin quantum was chosen too long, and what would happen if it was too short?

37) Mention four CPU scheduling events when scheduling can occur during the life cycle of a process?

CLO 6: (Marks 3) Section 5: Memory Management Each has 1 Marks

38) For the Contiguous Memory Allocation with variable size partitions mention the names of the registers used inside the Memory Management UNIT and their rolls?

base & limit registers:

39) In a paging system assume it takes 20 nsec to search TLB and 100 nsec to access memory. 98% of the time we are lucky and get the pages numbers in the TLB. Calculate the effective memory access time.

$$10.98 \times 120 + 0.02 \times 220$$

40) A pure paging system uses 16-bit address and 1K pages. The following shows the page tables of Process 1. Translate the logical address 2048 of Process Fill your answers into the table below.
[1 Marks]

Process 1

0	2
4	3
5	8
2	0

Process	Address	Page number	offset	Physical Address
---------	---------	-------------	--------	------------------

Process 1	2048	5	0	2 x 1024
-----------	------	---	---	----------

CLO 8: (Marks 6) Section 6: Disk Management

41) Hard disk block size is an important issue in file system design. What are the advantages and disadvantages of choosing large or small block size [1 Marks]

42) Assume that the block size is 1024 bytes, fill in the following spaces [2 Marks]

1. The physical Address of byte number 1045 in a file stored using contiguous allocation is _____. Keep in mind that this file is stored in several block starting from block number 19 and end at block 24.

22nd byte of 20th block

2. The physical Address of byte number 1045 in a file stored using the following FAT is _____. Keep in mind that the starting entry for this file is block 4.

22nd byte of 7th block

		8		7			2	-1	
0	1	2	3	4	5	6	7	8	9

43) For A Linux file system with i-node of 13 pointers and uses a block of size 8 Kbytes, consider a file of size 84 Kbytes. For such file, fill in the space in the following:

[3 Marks]

- i. The file occupies a number of9..... blocks and cause internal fragmentation of size4.....Kbytes.
- ii. If you want to read the byte number 2010, you will need ___1___ disk access but if you want to read the byte 83000 you will need ___2___ disk access.
- iii. If you want to increase the size of this file, the max size of this file will be ___8TB___ and the last byte can be read using ___4___ disk access.

$$\boxed{84 \text{ Kb} / 8 \text{ kbytes} = 11}$$